Anjali Singh, PhD |

anjali.singh@ischool.utexas.edu • anjalisinghphd.github.io

Human-centered systems researcher and designer with 9 years of experience conducting research in human-AI interaction, learning, information seeking and human behavior. Led research across industry and academia to develop AI-powered systems for learning, search, and programming, through close collaboration with cross-functional teams. Experienced driving product and design decisions through empirical research, experimentation, behavioral analysis and stakeholder engagament.

Research Areas: Human-AI Interaction, Learning & Education, Information Seeking, Cognitive Science

Education

School of Information — **University of Michigan, Ann Arbor**
Ph.D. in Information — *2019–2024*

Mathematics and Computing — **Indian Institute of Technology, Delhi**
Integrated Bachelor and Master of Technology — *2012–2017*

Research and Work Experience

University of Texas at Austin | Postdoctoral Research Fellow
Enhancing Learning & Information Seeking with Generative AI — *Sept'24–Current*

- **Led the design, prototyping, and evaluation** of an **interactive GenAI-powered search and note-taking system**, that provides metacognitive support during information-seeking, helping users critically engage with information [Demo].
- **Developed novel behavioral metrics** including persistent inquiry and topic coverage using interaction logs (chat history, notes, and clickstream data) to **characterize user workflows** and **evaluate learning and exploration behaviors**.
- Designed and conducted Wizard-of-Oz studies, user interviews, and behavioral analyses to understand how users engage with AI for searching, interact with AI-generated guidance, and inform design decisions to enhance GenAI search.
- **Increased exploration breadth** by 88%, **sustained inquiry behaviors** by 60%, and **user questioning** by 46%, compared to baseline system, demonstrating improved engagement and deeper information exploration.
- Applied **LLM-assisted qualitative coding, summarization and thematic analysis** to synthesize user data and outcomes and identify patterns in reasoning, trust, and engagement.

Responsible and Scalable AI-Integration in Higher Education — *Sept'24–Current*

- Conducted usability testing, interviews, and stakeholder research with instructors and administrators to identify user needs, product requirements and deployment considerations for a **campus-wide AI tutor platform at UT Austin**.
- Generated actionable **design recommendations spanning pedagogy, transparency, privacy, and user experience** that informed platform design and development.
- Designed & supported user studies evaluating impact of **multimodal GenAI tutors** on students' learning outcomes, engagement and experience, focusing on **trustworthiness, safety and accuracy**; Led to 0.61σ **boost in student scores**.
- Identified **opportunities for assessment redesign** in the AI era via instructor interviews, surveys, & co-design sessions.

Microsoft, Redmond + University of Michigan (U-M) | UX Research Intern
Understanding Data Science Learning and Feedback Workflows — *Jan'22–July'22*

- **Led design and deployment** of personalized feedback generation tool for data science courses at U-M, in partnership with 10-member cross-functional team of researchers, product managers, and developers across Microsoft and U-M.
- **Facilitated data sharing between U-M and Microsoft**. Identified common misconceptions and core competencies on which novice data scientists and learners require support through log analysis and qualitative code analysis.
- **Collected and analyzed user feedback**, including perceived helpfulness of feedback tool and level of learning support.
- **Disseminated findings with Microsoft leadership**, informing future directions for data science developer tools.

University of Michigan | Graduate Researcher
Learning Experience Design, Behavioral Analysis, Aligning with Student Motivations — *Jan'20–Aug'24*

- Designed and led **large-scale field experiments** involving 3800+ learners to evaluate learning technologies promoting critical thinking, agency, and active learner-AI engagement, across informal and formal learning contexts.
- Generated **insights into motivation, participation, and effort** that informed the design of scalable learning systems.
- Developed **AI-integrated personalized learning experiences** using **LLM prompting** leveraging data from past cohorts; **Evaluated impact using metrics** such as retention, performance, learning outcomes and perceived value.
- Applied machine learning and behavioral clustering techniques to **identify patterns in user interaction and learning strategies using ebook log-data**, obtaining 3 distinct behavioral groups.

IBM Research, Bengaluru | Research Software Engineer
AI for Social Good—Early Chilhood Learning & Civic Engagement *Jul'17–May'19*

- *Visual Assessment Generation.* Generated insights for picture-based assessment development through interviews with instructors; Leveraged **image captioning, open information extraction, and neural embedding based semantic similarity matching** to auto-generate visual assessments using knowledge base and image corpus.
- *Civic Issue Reporting.* Conducted user research with Indian citizens to identify barriers and preferences for reporting civic issues. **Mined 1.5k images** with civic issues and associated captions for unsupervised training of deep learning model to automatically detect civic issues from images, **improving accessibility and effectiveness of civic reporting tools**.

Technical Skills & Interests

Quantitative Research Methods: A/B Testing & Experimentation, Statistical Modeling & Analysis (e.g., Regression, Cluster Analysis), Causal Inference, Behavioral Analysis, Log Analysis, Text Analysis, Applied AI/ML, Data Visualization

Qualitative Research Methods: Interviews, Surveys, User Studies, Usability Testing, Wizard-of-Oz studies, Contextual Inquiry, Focus Groups, Diary Studies, Co-design, Thematic Analysis

Technical: Python (NumPy, Pandas, SciPy, Matplotlib, Scikit-Learn, NLTK), R, SQL, React, Prompt Engineering, LLM APIs, AI-Assisted Prototyping

Tools & Platforms: Qualtrics, Prolific, Canva, Figma, Atlas.ti, MAXQDA, Jupyter, VSCode, RStudio, Github, Claude Code

Product & Collaboration: Research Strategy & Leadership, Stakeholder Engagement, Cross-functional Collaboration

Selected Awards

Best Paper Awards: ACM Learning@Scale Conference, 2021 | Learning Analytics and Knowledge Conference, 2024

Awarded $200,000 Funding for Doctoral Research: Microsoft, 2022

Summer Undergraduate Research Award: Indian Institute of Technology Delhi, 2014

Gold Medal: Indian Physics Olympiad (InPhO), Awarded to Top 35 Indian high-school students as per InPhO exam, 2012

Selected Publications **[Google Scholar, 515+ citations, h-index: 9, 15+ publications]**

MetaCues: Enabling Critical Engagement with Generative AI for Information Seeking and Sensemaking. A. Singh et al. *arXiv preprint*. 2026 [Paper]

Impact of Multimodal and Conversational AI on Learning Outcomes and Experience. K. Taneja, A. Singh, A. K. Goel. *International Conference on Artificial Intelligence in Education (AIED)*. 2026 [Paper]

Toward Responsible and Scalable Integration of Course-Specific AI Tutors: Instructor Experiences with a Campus-Wide Platform. G. Ko, H. K. Lee, A. Singh, L. Boddy, K. Ford, and E. Huff Jr. *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. [Paper]

Enhancing Critical Thinking in Generative AI Search with Metacognitive Prompts. A. Singh et al. *Proceedings of the Association for Information Science and Technology (ASIS&T)*. 2025 [Paper]

What's In It for the Learners? Evidence from a Randomized Field Experiment on Learnersourcing Questions in a MOOC. A. Singh. et al. *Proceedings of the Eighth ACM Conference on Learning@Scale (L@S)*. 2021. Best Paper Award [Paper]

Research Outreach & Community Leadership

Invited Talk: Designing Technologies for Critical and Active Learning in the Age of AI. *EPFL*, 2025

Organised Workshop: Tools for Thought: Understanding, Protecting, and Augmenting Human Cognition with Generative AI–From Vision to Implementation. *CHI 2026 Conference*

Organised Workshop: Empowering Education with LLMs - the Next-Gen Interface and Content Generation. *Artificial Intelligence in Education Conference*, 2023

Invited Talk: Feedback Generation using Human-AI Collaboration for Introductory Data Science. *Microsoft*, 2021

Leadership & Mentoring

2020–Current: **Program Committee Member/Reviewer.** *Conferences:* CHI ('25,'24,'23), Learning@Scale ('25,'26), Learning Analytics and Knowledge'26, SIGCSE ('24,'23), Educational Data Mining Conference ('21); *Journals*: Computers & Education, Human-Computer Interaction, Journal of Learning Analytics

2018–Current: **Mentored** 5 undergraduate and 6 graduate students at the University of Michigan, UT Austin, and Georgia Tech and 3 research fellows and interns at IBM Research and Microsoft

2021: **Mentored** 2 teams in Educational Data Mining track, **LearnLab Summer School, Carnegie Mellon University**

2018: **Organising team member** for **Hackathon on AI for Social Good** at IBM Research Labs India